PAPER ID-310680

Roll No: | | | | | | | | | | | | | |

# BTECH
## (SEM V) THEORY EXAMINATION 2024-25
## DESIGN AND ANALYSIS OF ALGORITHM

**TIME: 3 HRS** **M.MARKS: 70**

**Note:** Attempt all Sections. In case of any missing data; choose suitably.

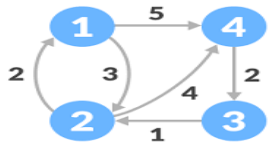
### SECTION A

**1. Attempt *all* questions in brief.** **2 x 07 = 14**

| Q no. | Question | CO | Level |
|---|---|---|---|
| a. | With example define algorithm. List few algorithm design techniques. | 1 | K1 |
| b. | Briefly discuss the basic steps taken to design an algorithm. | 1 | K1 |
| c. | Derive the time complexity of Heap Sort. | 2 | K2 |
| d. | List the properties of Binomial Heap | 2 | K1 |
| e. | With a suitable example explain the concept of Convex –Hull Problem | 3 | K2 |
| f. | With a suitable example explain "Branch and Bound". | 4 | K2 |
| g. | Describe "Randomized algorithms". List few randomized algorithms. | 5 | K2 |

### SECTION B

**2. Attempt any *three* of the following:** **07 x 3 = 21**

| | | | |
|---|---|---|---|
| a. | Illustrate the operation of Merge –Sort on array A= (38, 27, 43, 3, 9, 82, 10). Also drive the time complexity of Merge Sort. | 1 | K3 |
| b. | Define Binomial Heap. Write an algorithm for union of two binomial heaps. Also take a suitable example which clearly illustrates merging operation of two binomial heaps. | 2 | K2 |
| c. | Apply the greedy single source shortest path algorithm on the graph given below.   | 3 | K3 |
| d. | Write Floyd's and Warshal's algorithm to find all pair shortest path in a graph. Discuss its time complexity. | 4 | K2 |
| e. | Explain Vertex Cover Problem. Solve vertex cover problem using approximation algorithm | 5 | K2 |

### SECTION C

**3. Attempt any *one* part of the following:** **07 x 1 = 07**

| | | | |
|---|---|---|---|
| a. | Write Quick –Sort partition algorithm. Drive best and worst case time complexity of quick sort. | 1 | K2 |
| b. | Find out Upper, Lower and Average bounds for the function f (n) = 3n+2 | 1 | K3 |

**4. Attempt any *one* part of the following:** **07 x 1 = 07**

| | | | |
|---|---|---|---|
| a. | Insert the following string in the initially empty tries: DOG, DONE, CAT, CAN, RIGHT, DO, JUG, DAA, CA, CAME. Also make a compress tries of it. | 2 | K3 |
| b. | Design a Binomial Heap for the following A. A= {7, 2, 4, 17, 1, 11, 6, 8, 15, 10, 20} | 2 | K3 |



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| **5.** | **Attempt any *one* part of the following:** | **07 x 1 = 07** | |
|---|---|---|---|
| a. | Write and explain the Kruskal's algorithm to find Minimum Spanning Tree of a graph with a suitable example. | 3 | K2 |
| b. | Find the optimal solution of the fractional Knapsack problem with n=7 and the knapsack capacity of m=15. The profits and weights of the items are given below.<br>Objects: 1 2 3 4 5 6 7<br>Profit (P): 5 10 15 7 8 9 4<br>Weight (w): 1 3 5 4 1 3 2 | 3 | K3 |

| **6.** | **Attempt any *one* part of the following:** | **07 x 1 = 07** | |
|---|---|---|---|
| a. | Illustrate the N-queens problem? Draw "State Space Tree" for 4 queen's problem using backtracking. | 4 | K3 |
| b. | Find the optimal solution to the 0/1 Knapsack instances with n=4 and Knapsack capacity m=8 where profits and weights as follows : P={1, 2, 5,6} and W={2, 3, 4, 5} | 4 | K2 |

| **7.** | **Attempt any *one* part of the following:** | **07 x 1 = 07** | |
|---|---|---|---|
| a. | Explain P, NP, NP –Complete and NP-Hard complexity classes. How they are related to each other. | 5 | K2 |
| b. | Write Knuth-Morris-Pratt string matching algorithm. Take a suitable example Compute the prefix function π for the pattern ababbabbabbababbabb when the alphabet is ∑ = {a, b}. | 5 | K3 |

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